

Graph Theory Problem Set

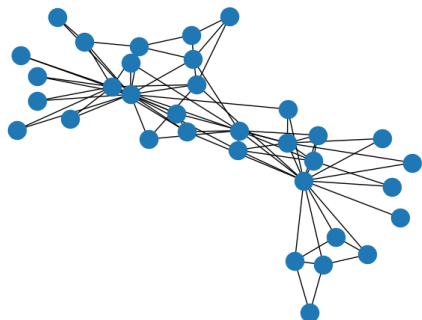


Figure 1: Karate club network.

Lots of useful networks have been collected by the **The KONECT Project** (konect.cc) . The graph in Figure 1 is known as **Zachary's karate club**. It is a very famous social network describing a university karate club which split due to a conflict between its members. The karate instructor, Mr. Hi, is labelled 1 and the club president, John A., is labelled 34. A conflict arose over increasing the lesson fees with one faction (pro-increase) lead by Mr. Hi and the other faction lead by John A. The rest of the nodes represent other club members and the links indicate if those members interacted outside of karate lessons and club meetings (i.e. friendships). The network was recorded just before the club split into two separate clubs. The karate club paper was reasonably well known in its own right but became especially famous with the growth of network science where it is a small, interesting and easily illustrated network that is often used as an example.

Problem 1: Download the karate club data from the KONECT database and use the python package `networkx` to plot the karate club network, get its degree sequence, find the degree of Mr. Hi and plot the degree distribution.

Problem 2: Repeat question 1 using `igraph`.

Problem 3: Repeat question 1 using `gephi`.

Note: I will only use **networkx** for the remainder of the course, but you may like some of the other options. These three are probably the most popular libraries, but not the only ones, and you're free to use whatever you want. An interesting one I have not tried much is Cytoscape.

Problem 4: The simplest bipartite left-projection is

$$A_{ij} = \sum_k^M B_{ik} B_{jk}$$

(i) Write this as a matrix equation.

(ii) Write an expression for $2L = \sum_{ij} A_{ij}$, the number of edges in the projected network, in terms the degrees of the projected nodes, r_k .

(iii) How many edges does a right hand node with degree r_k contribute to the projection?

(iv) A graph is called **k-regular** if all the degrees are equal to k . Assume that all the degrees $r_i = r$ and compute $\frac{F}{2L}$ in terms of r where $F = \sum_{ik}^{NM} B_{ik}$

(v) A **hyperbolic projection** is $A = BWB^T$ where W is a diagonal matrix with diagonal entries $\frac{1}{r_k}$. Compute $\frac{F}{2L}$ using this projection.

Problem 5:

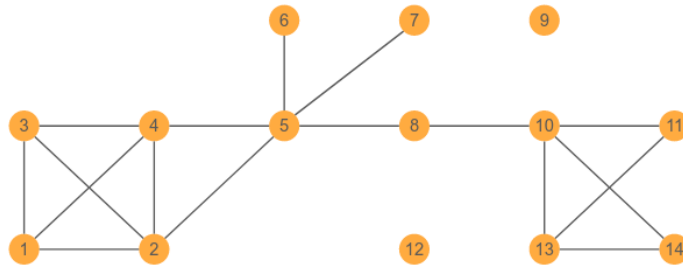


Figure 2: A network with some k-cores.

A **k-core** is the largest subgraph H with the property that $d(i) \geq k$ for all $i \in H$. Find the 0-, 1-, 2- and 3-cores of the graph in Figure 2. An algorithm to find k-cores is

1. Compute the degrees of vertices and sort them to get a list of nodes V
2. For $v \in V$:
 - (a) $\text{core}(v) = d(v)$
 - (b) Cut the link to every neighbour u of v where $d(u) > d(v)$, in which case $d(u) \rightarrow d(u) - 1$.
 - (c) Remove v from V , resort it and start from the beginning.

You can do the above network by hand, but if you have time you can try to implement the algorithm in code.